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Analysis of Early Brain Development via Longitudinal MRI Registration Using Stationary Velocity Fields

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Context

→ In gyrencephalic mammals, brain morphology exhibits rapid, non-linear transformations during fetal and postnatal period, characterised by the cortical folding that deforms the initially smooth surface of the brain to a highly folded one. Understanding these transformations is essential to studying brain development and its underlying mechanisms.

→ Stationary velocity field (SVF)-based registration [1] provides a computationally efficient framework for modelling diffeomorphic deformations by parameterising smooth, invertible transformations via stationary velocity fields.

Questions :

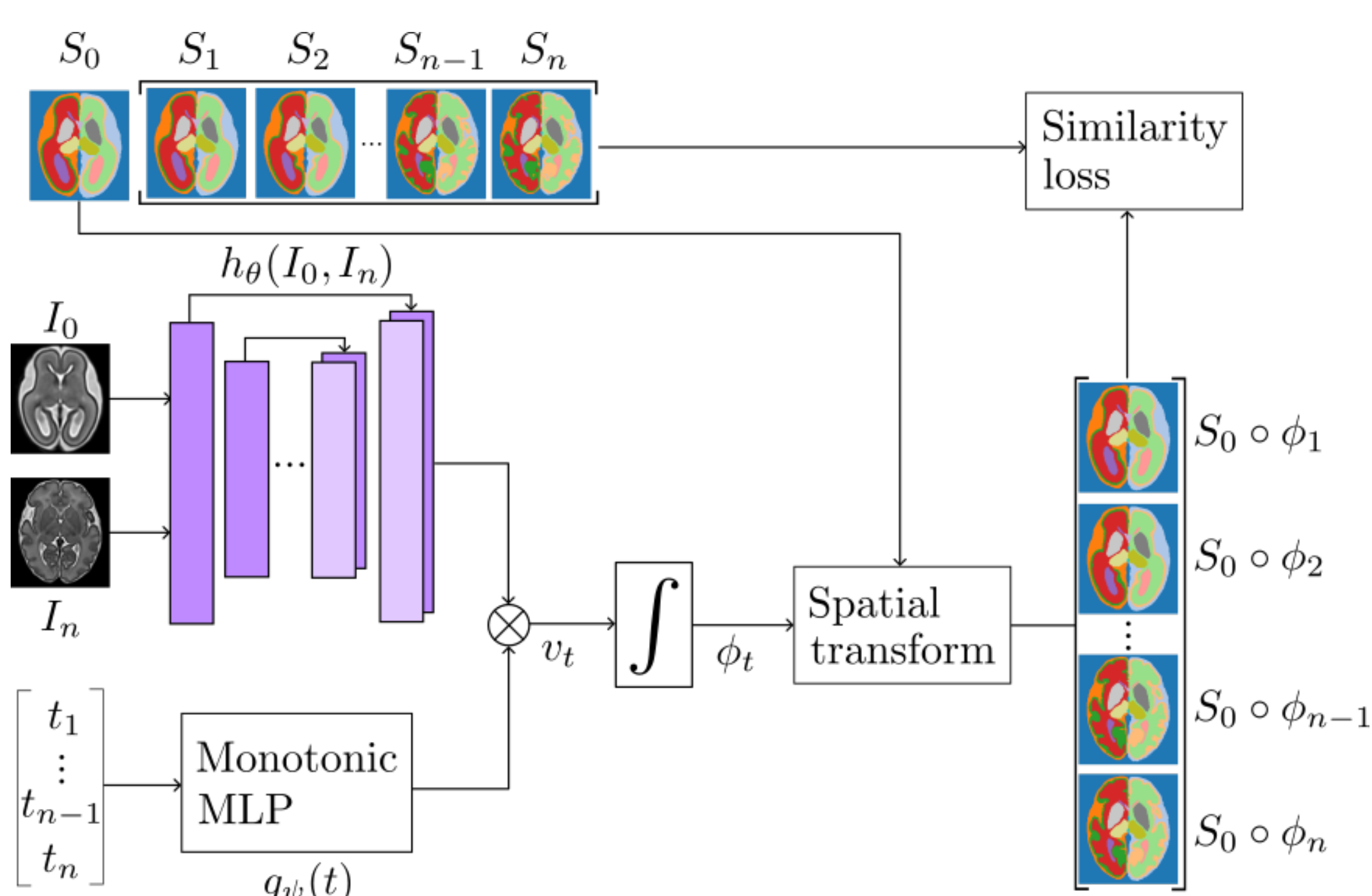
- How are efficient SVF-based registration methods to capture the brain deformations that occur during gyrification?
- How can the integration of the MRI longitudinal sequence improve the capture of the non-linear developmental dynamic?

Method

SVF-based longitudinal registration – Modeling a continuous deformation field ϕ from a longitudinal image sequence $\mathcal{S} = \{I_0, I_1, \dots, I_n\}$, allowing I_0 to be warped to any desired time point.

- SVF framework** – A neural network predicts a SVF between the source image I_0 and the target image I_n . The SVF is integrated to obtain a deformation field ϕ , which is then used to warp the source image and generate the predicted image.
- SVF weighting** – The SVF is modulated by a scalar factor s , defined either linearly ($s = t$) or via a multi-layer perceptron ($s = g(t)$), to compute an adapted data-driven velocity field v at any time step.
- Network optimisation** – Parameters are learned via a data-driven self-supervised objective defined across all time points of the longitudinal MRI sequence. A similarity term between the aligned baseline segmentation map S_0 (associated with I_0) and the subsequent segmentation maps ($S_1, \dots, S_n$) is employed in the optimisation problem:

$$\theta^* = \arg_{\theta} \min \sum_{k=1}^n \mathcal{L}_{sim}(S_0 \circ \phi_k, S_k) + \lambda \mathcal{L}_{reg}(\phi_k)$$



Evaluation of the effectiveness of SVF-based registration approaches

- Multi-species analysis** – Applied to three longitudinal atlases of early mammalian brain development: human [2], ferret [3], and rhesus macaque [4].
- Different registration paradigms** – Pairwise (SVF-Pair), Pairwise with interpolation (SVF-Int), and longitudinal approaches (HH [5] and our investigated methods without non-linear SVF weighting (SVF-LIN) and with (SVF-MLP).
- Metrics** – Quantitative evaluation based on segmentation maps overlap (Dice), and the gyrification index [6].

Results

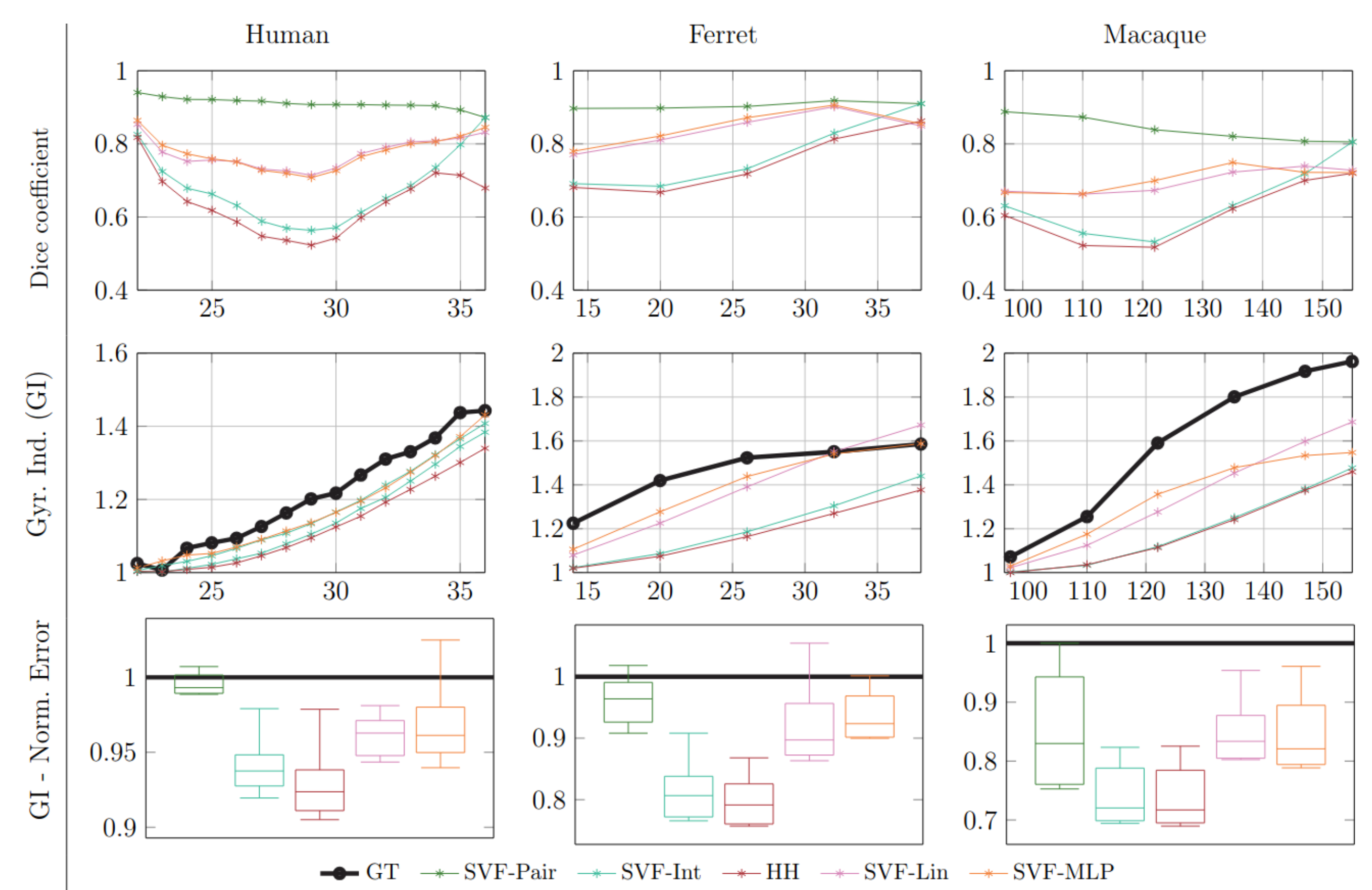


Figure 1. Quantitative evaluation of the Dice coefficient for the cortex (first row) and the gyrification index (second and third rows). The second row shows the temporal evolution of the gyrification index. The third row illustrates the normalised error that measure the deviation from the reference gyrification trajectory.

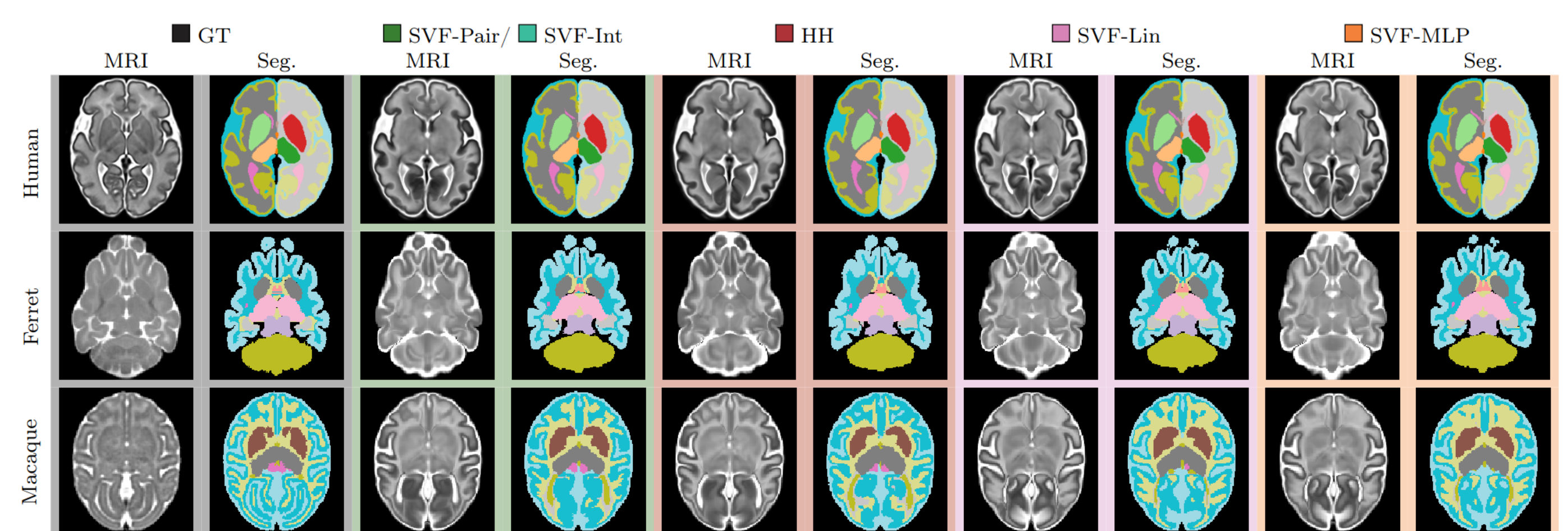


Figure 2. Visual evaluation of the compared SVF-based image registration methods. For each longitudinal MRI sequence, an axial slice from the first MRI and its segmentation map (smooth brain) is registered to the last time point of the sequence.

Conclusion

- SVF-based longitudinal methods that leverage the entire longitudinal sequence reveal lower variance in the gyrification index and better capture of the developmental dynamics.
- SVF-based registration methods appear to only partially capture brain deformations associated with gyrification, as evidenced by a shared underestimation of cortical deformations.

References

- V. Arsigny et al., "A Log-Euclidean Framework for Statistics on Diffeomorphisms," in MICCAI, 2006.
- A. Schuh et al., "Unbiased construction of a temporally consistent morphological atlas of neonatal brain development," en, *BioRxiv*, 2018.
- K. E. Garcia et al., "Longitudinal MRI of the developing ferret brain reveals regional variations in timing and rate of growth," en, *Cerebral Cortex*, 2024.
- J. A. Karpf et al., "Characterization of normative fetal rhesus macaque brain development with magnetic resonance imaging," *Imaging Neuroscience*, 2025.
- M. Hadji-Hamou et al., "Longitudinal analysis of image time series with diffeomorphic deformations: A computational framework based on stationary velocity fields," *Frontiers in Neuroscience*, 2016.
- K. Zilles et al., "The human pattern of gyrification in the cerebral cortex," *Anatomy and Embryology*, 1988.

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